Luise Ge (Haijing)

g.luise@wustl.edu
luisegehaijing.github.io
Google Scholar

RESEARCH INTERESTS

I work at the intersection of **AI and economics**. I am passionate about bringing theoretical depth to empirical questions that affect many people, such as how AI systems should aggregate the preferences of a diverse population. My main interests are therefore **multi-agent systems**, **computational social choice**, and **AI alignment**. Because of these problems, I am also motivated to dive deeper into **algorithm and mechanism design**, **learning theory** and **formal reasoning**.

EDUCATION

Washington University in St. Louis — St. Louis, MO
Ph.D. in Computer Science, GPA 4.00/4.00 — 2023 – present
- Advisors: Prof. Yevgeniy Vorobeychik and Prof. Gregory Kehne

Imperial College London — London, UK
M.Sc. in Pure Mathematics, with Distinction — 2022 – 2023
- Advisor: Prof. Paolo Cascini

University of Edinburgh — Edinburgh, UK
B.Sc. (Hons) in Artificial Intelligence and Mathematics, First Class — 2018 – 2022
- Honours thesis: *Computing Competitive Equilibria for Fair Allocation of Chores* (Outstanding Honours Project)

HONORS & AWARDS

Outstanding Paper Award, ACL (top 0.4% of accepted papers) — 2026
Best Research Paper Award shortlist, ECML PKDD — 2026
Oral presentation, recommended for award, AAAI — 2026
Spotlight presentation, NeurIPS (top 2.1% of submissions) — 2024
Cox Fellowship, Washington University in St. Louis — 2023
Best Final Year Performance by a Woman Student, School of Informatics, Univ. of Edinburgh — 2022
Outstanding Honours Project, School of Informatics, Univ. of Edinburgh — 2022

PUBLICATIONS

In computer science, peer-reviewed conference proceedings are the main archival venue. (α-β) marks alphabetical author order, the convention in theoretical computer science. Awards in red.

PEER-REVIEWED CONFERENCE PAPERS

[C1] (α-β) **L. Ge**, D. Halpern, G. Kehne, Y. Vorobeychik. "Linear Social Choice with Few Queries: A Moment-Based Approach." *38th ACM-SIAM Symposium on Discrete Algorithms* (SODA), 2027. Previously presented at ESIF AI+ML 2026.

[C2] **L. Ge**, Y. Zhang, Y. Vorobeychik. "Mind the (DH) Gap! A Contrast in Risky Choices Between Reasoning and Conversational LLMs." *64th Annual Meeting of the Association for Computational Linguistics* (ACL), 2026. Oral; Outstanding Paper Award.

[C3] (α-β) **L. Ge**, G. Kehne, Y. Vorobeychik. "Optimized Distortion in Linear Social Choice." *40th AAAI Conference on Artificial Intelligence* (AAAI), 2026. Oral; recommended for award.

[C4] (α-β) **L. Ge**, B. Juba, K. Nilsson, A. Shao. "Lifted Relational Probabilistic Inference via Implicit Learning." *42nd Conference on Uncertainty in Artificial Intelligence* (UAI), 2026.

[C5] B. Rachmut, **L. Ge**, N. Zhang, W. Yeoh, Y. Vorobeychik. "COSMOS: Model-Agnostic Personalized Federated Learning with Clustered Server Models and Pseudo-Label-Only Communication." *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases* (ECML PKDD), 2026. Best Research Paper Award shortlist.

[C6] **L. Ge**, B. Juba. "Query-Driven Learning for Lifted Reasoning." *6th International Joint Conference on Learning & Reasoning* (IJCLR), 2026.

[C7] **L. Ge**, M. Lanier, A. Sarkar, B. Guresti, C. Zhang, Y. Vorobeychik. "Learning Policy Committees for Effective Personalization in MDPs with Diverse Tasks." *42nd International Conference on Machine Learning* (ICML), 2025.

[C8] (α-β) **L. Ge**, B. Juba, K. Nilsson. "Polynomial-Time Relational Probabilistic Inference in Open Universes." *34th International Joint Conference on Artificial Intelligence* (IJCAI), 2025.

[C9] (α-β) **L. Ge**, D. Halpern, E. Micha, A. D. Procaccia, I. Shapira, Y. Vorobeychik, J. Wu. "Axioms for AI Alignment from Human Feedback." *38th Conference on Neural Information Processing Systems* (NeurIPS), 2024. Spotlight (top 2.1% of submissions).

WORKSHOP PAPERS & ABSTRACTS

[W1] **L. Ge**, B. Juba, Y. Vorobeychik. "Learning Linear Utility Functions From Pairwise Comparison Queries." Social Choice and Learning Algorithms (SCaLA) Workshop at *23rd International Conference on Autonomous Agents and Multiagent Systems* (AAMAS), 2024; one-page abstract at *8th International Conference on Algorithmic Decision Theory* (ADT), 2024.

WORK IN PROGRESS

[U1] **L. Ge**, G. Kehne, Y. Vorobeychik. "Utilitarian or Beyond? Welfare Modeling for Pluralistic Alignment Needs to Cost More Data."

[U2] **L. Ge**, E. Poole-Dayan. "Pluralistic AI Alignment: A Survey Bridging Theory and Practice."

[U3] M. Lanier, **L. Ge**, S. Kompella, Y. Vorobeychik. "Learned Neighbor Trust for Collaborative Deployment in Model-Agnostic Decentralized Learning."

TEACHING & MENTORING

Instructor & Teaching Assistant (mentored teaching), *AI and Society*, WashU — 2026
Demonstrator, *Cognitive Science*, Univ. of Edinburgh — 2021, 2022
Tutor, *Introduction to Algorithms & Data Structures*, Univ. of Edinburgh — 2021
Tutor, *Introduction to Computation & Logic*, Univ. of Edinburgh — 2020

Research mentee: Yongyan Zhang (undergraduate), co-author of [C2]; now M.S., Carnegie Mellon Univ.

SERVICE & OUTREACH

Conference reviewer: NeurIPS 2025, 2026; ICLR 2025; ECML PKDD 2026; AAMAS 2027; IASEAI 2027
Journal reviewer: *Journal of Artificial Intelligence Research* (JAIR), 2025

Graduate Ambassador, Washington University in St. Louis — 2024 – present
Mathematics Student Representative, Imperial College London — 2023
Edinburgh Award (Volunteering, 50+ hours), University of Edinburgh — 2020
Edinburgh Award (Change Agents), University of Edinburgh — 2019

PROJECTS

Synchria (synchria.xyz) — Independent project, 2026

- Investigating computational mechanisms to resolve search frictions and streamline human collaboration. Built an end-to-end matching architecture deployed at Washington University in St. Louis; funded by the Skandalaris Center and featured in campus media, with the long-term goal of scaling open-ecosystem human search.

SKILLS

Programming: Python, Java, MATLAB
Optimization: Gurobi, CVXPY
Machine Learning & Systems: PyTorch, Hugging Face, vLLM, multi-node/multi-GPU training on SLURM clusters, Google Cloud Run
Empirical Methods: Human-subjects experiment design, bootstrap inference, behavioral model fitting